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Specification For OED 13.3" EPD

Model NO.: OPM133A1

Prepared by	Checked by	Approved by
DR Yu	JQ Sun	XD Wang

Customer Approval

Customer	Approved by	Date of Approval

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	Version	Content	Date	Producer	
	A/0	New release	2024/01/15	Yu DR	
	A/1	Update the dimension of FPCA	2024/05/09	Yu DR	

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1 General Description

OPM133A1 is an Active Matrix Electrophoretic Display, with integrated circuits including source and gate drivers, can be used in portable electronic devices, such as E-book Reader. It has 13.3inch active area with 2200 x 1650 pixels, the module is capable to display white/black images at 2-16 gray levels(1-4 bits).

2 Features

- ◆ 2200 x 1650 display
- ◆ High contrast reflective / electrophoretic technology
- ◆ Wide viewing angle
- ◆ Ultra low power consumption
- ◆ Reflective mode
- ◆ Bi -stable display
- ◆ Commercial temperature range
- ◆ Landscape mode

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3 Pin Assignment

No.	Pin Name	Description
1	VGL	Negative power supply gate driver
2	NC	NO Connection
3	VGH	Positive power supply gate driver
4	NC	NO Connection
5	VDD	Digital power supply drivers
6	OEV	Output mode selection gate driver
7	CKV	Clock gate driver
8	SPV	Start pulse gate driver
9	NC	NO Connection
10	VCOM	Common voltage connection
11	NC	NO Connection
12	VCOM	Common voltage connection
13	NC	NO Connection
14	D0	Data signal source driver
15	D1	Data signal source driver
16	D2	Data signal source driver
17	D3	Data signal source driver
18	D4	Data signal source driver
19	D5	Data signal source driver
20	D6	Data signal source driver
21	D7	Data signal source driver
22	VSS	Ground
23	CKH	Clock source driver
24	VSS	Ground
25	D8	Data signal source driver
26	D9	Data signal source driver
27	D10	Data signal source driver
28	D11	Data signal source driver
29	D12	Data signal source driver
30	D13	Data signal source driver
31	D14	Data signal source driver
32	D15	Data signal source driver
33	SPH	Start pulse source driver
34	LEH	Latch enable source driver
35	OEH	Output enable source driver
36	NC	NO Connection
37	NC	NO Connection
38	VCOM	Common voltage connection
39	NC	NO Connection
40	NC	NO Connection
41	VSH	Positive power supply source driver

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	42	NC	NO Connection	
	43	VSL	Negative power supply source driver	
	44	NC	NO Connection	
	45	BORDER	Border connection	

4 Electrical Characteristics

4.1 Module DC Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Signal ground	VSS		-	0	-	V
Logic Voltage supply	VDD		3.0	3.3	3.6	V
	IVDD	VDD=3.3V	-	3.3	10	mA
Gate Positive supply	VGH		-	22	-	V
	IVGH	VGH=22V	-	1.3	5	mA
Gate Negative supply	VGL		-	-20	-	V
	IVGL	VGL=-20V	-0.5	-1.7	-	mA
Source Positive supply	VPOS		14.6	15	15.4	V
	IPOS	VPOS=15V	-	3	400	mA
Source Negative supply	VNEG		-15.4	-15	-14.6	V
	INEG	VNEG=-15V	-400	-3	-	mA
Asymmetry source	VASYM	VPOS+VNEG	-800	0	800	mV
Common voltage	VCOM		-3.0	-2.10	-1.5	V
	ICOM		-	0.3	0.5	mA
Standby power module	PSTBY		-	163.5	-	mW
Typical power module	PTYP		-	10.9	-	mW
Operating temperature			0		50	℃
Storage temperature			-25	-	70	℃

Notes:

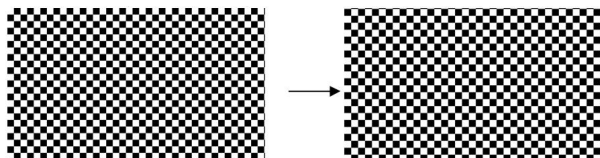
1. The maximum power and maximum current are specified for the worst case power consumption (Note 4-1).
2. The typical power is measured when “typical images” are displayed (Note 4-2).
3. The standby power is the consumed power when the module controller is in standby mode.

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4. The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by OED.

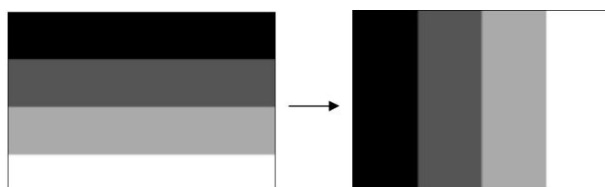
Note 4-1

The maximum power consumption



Note 4-2

The typical power consumption



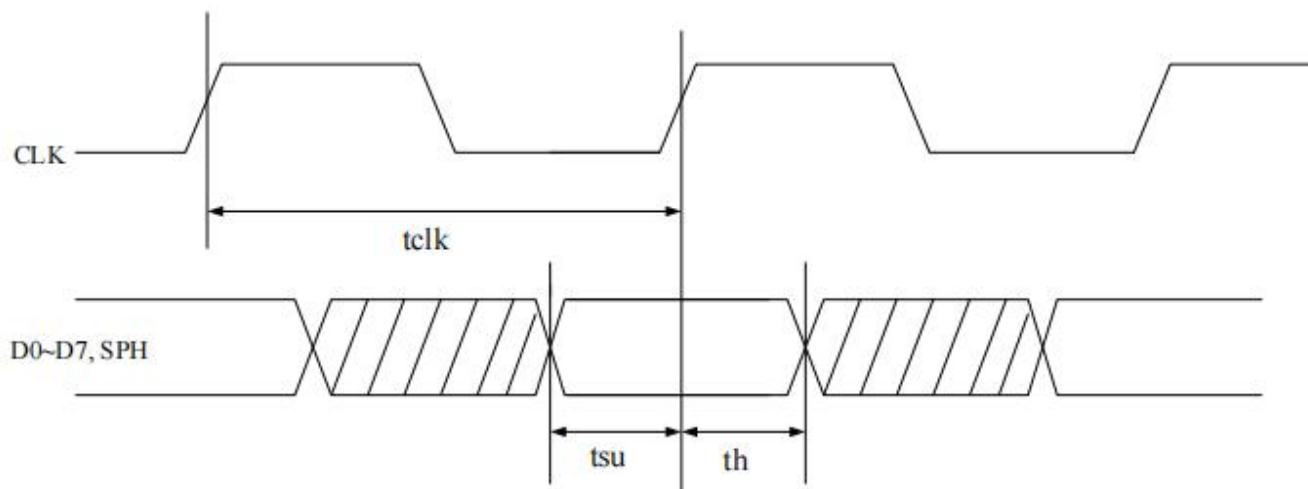
4.2 Module AC characteristics

Note: VDD = 3.0V to 3.6V, unless otherwise specified.

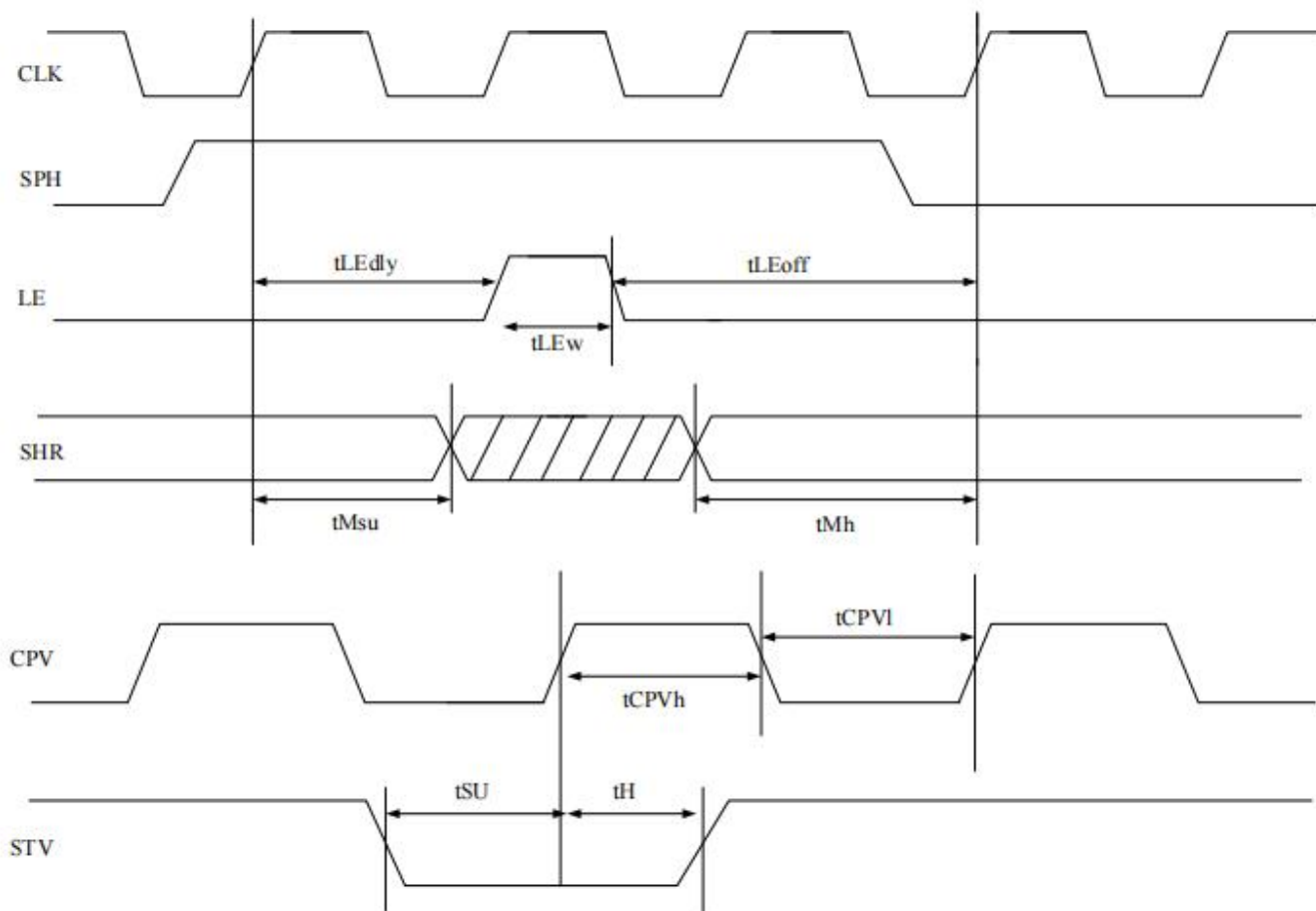
Parameter	Symbol	Min.	Typ.	Max.	Unit	App Pin
Clock frequency	fckv	-	-	200	kHz	CPV
Clock CPV high time	tCPVh	0.5	-	-	us	
Clock CPV low time	tCPVl	0.5	-	-	us	
Data setup time	tSU	100	-	-	ns	CPV STV
Data hold time	tH	100	-	-	ns	
Clock CLK cycle time	tclk	16.7	20	-	ns	Below table
D0 .. D15, SPH setup time	tsu	4	-	-	ns	
D0 .. D15, SPH hold time	th	4	-	-	ns	
LE on delay time	tLEdly	40	-	-	ns	
LE high-level pulse width	tLEw	40	-	-	ns	
LE off delay time	tLEoff	200	-	-	ns	
SHR setup time	tMsu	10	-	-	ns	
SHR hold time	tMh	10	-	-	ns	

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Clock & Data Timing



Output Latch/Control Signals



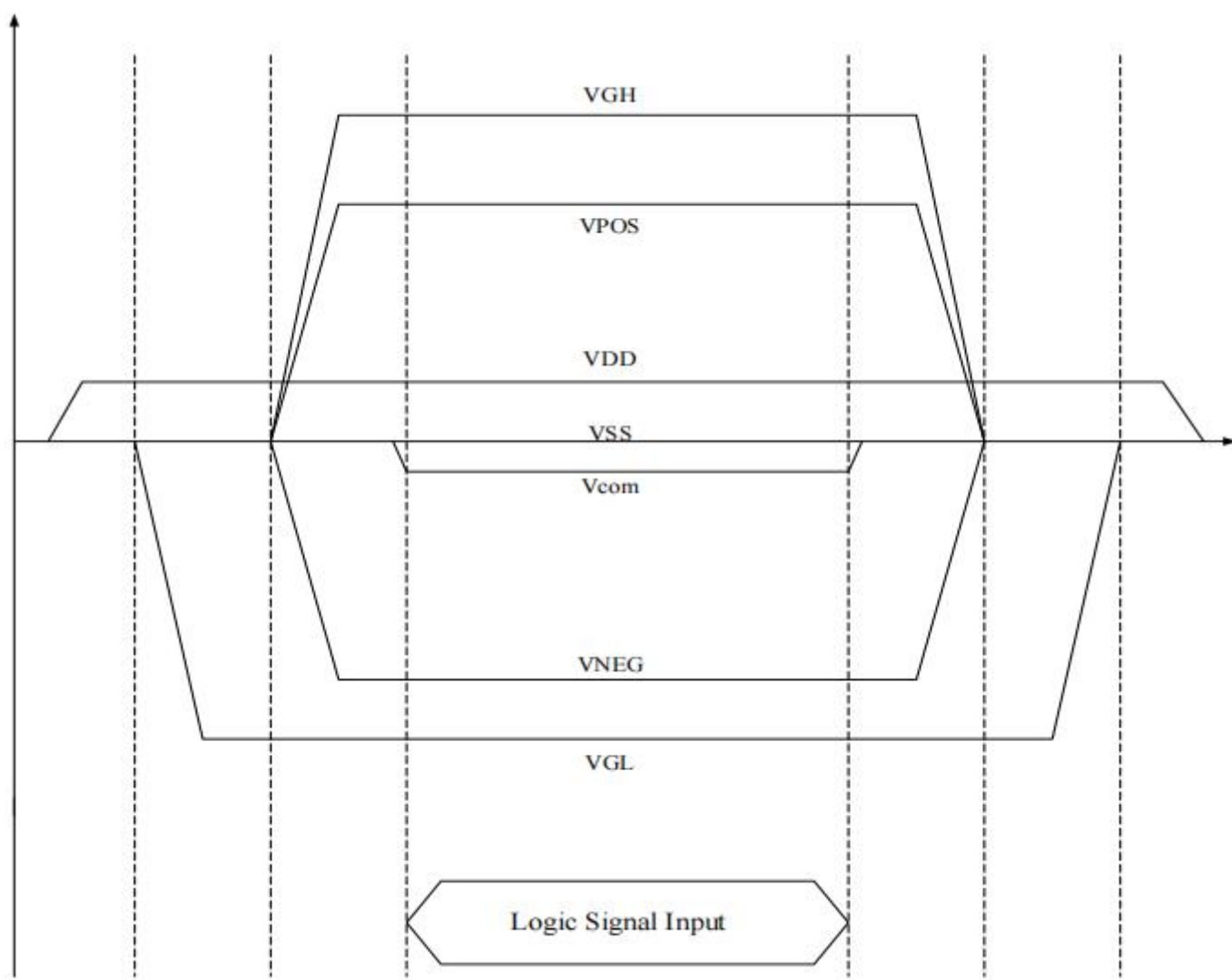
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5 Power On/Off Sequence

To prevent the device from damage due to latch up, the power on/off sequence shown below must be followed.

When power on: VDD -> VGL -> VNEG/VGH/VPOS -> Vcom

When power off: Vcom -> VNEG/VGH/VPOS -> VGL -> VDD



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6 Tcon Timing

SDCK(MHz)	46	Resolution 1650(H)*2200(V)				
Pixels Per SDCK	8					
Line Parameter(SDCK)	LSL	LBL	LDL	LEL	GDCK_STA	LGONL
	16	17	275	15	9	284
Line Parameter(us)	LSL	LBL	LDL	LEL	GDCK_STA	LGONL
	0.35	0.37	5.98	0.33	0.20	6.17
Frame Parameter(SDCK)	FSL	FBL	FDL	FEL	-	FR(Hz)
	1	4	1650	14	-	85
Frame Parameter(us)	FSL	FBL	FDL	FEL	-	-
	7.02	28.09	11585.87	98.3	-	-

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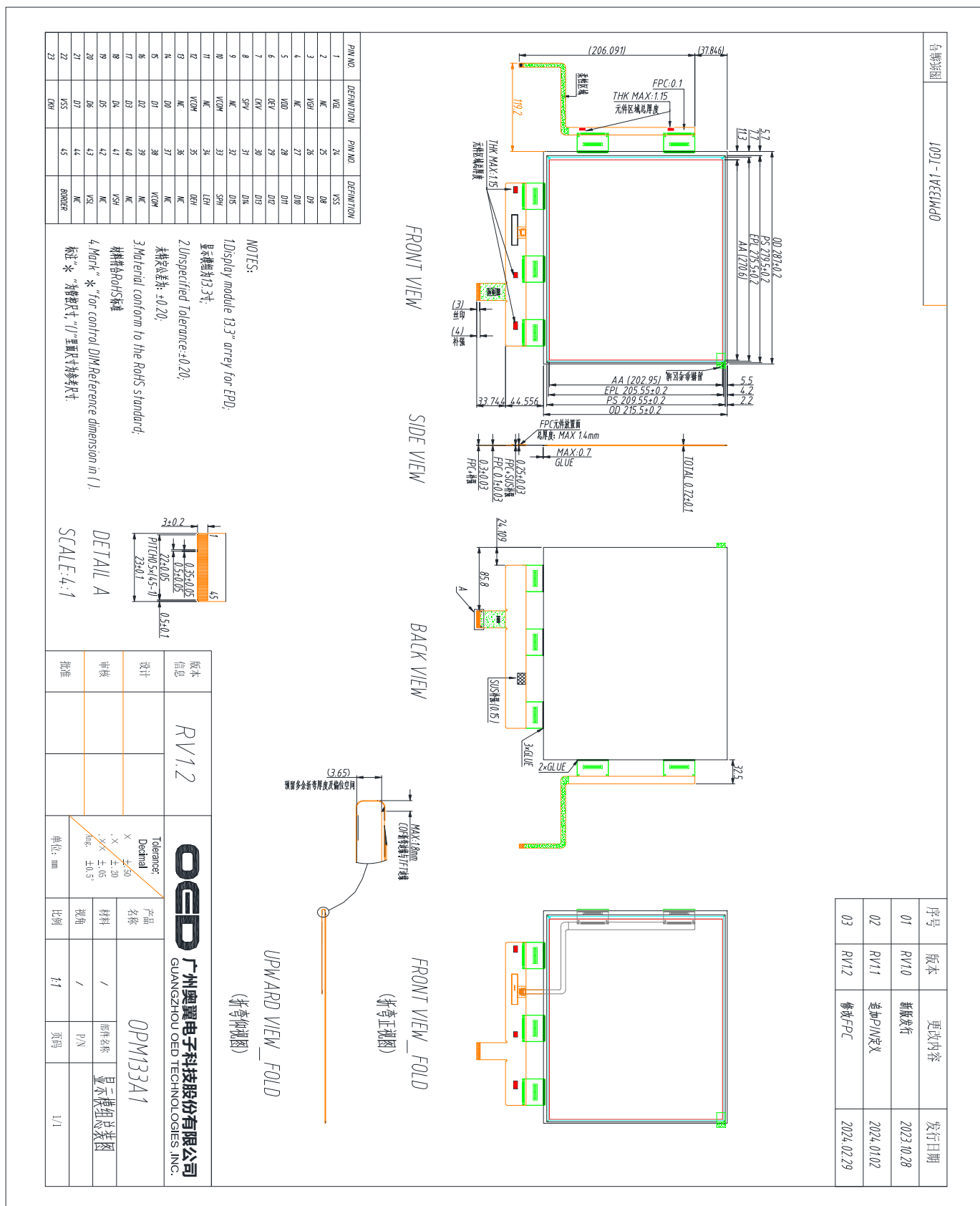
7 Mechanical Specifications

7.1 Dimension

PARAMETER	VALUE	UNIT	Remark
Display Resolution	2200 × 1650	dots	
Active Area Dimensions			
Horizontal	270.6	mm	
Vertical	202.95	mm	
Screen Size	13.3	Inch	
Pixel Pitch			
Horizontal	0.123	mm	
Vertical	0.123	mm	
Pixel Configuration	Square		
Overall dimensions			
Width	287	mm	
Height	215.5	mm	
Thickness	0.72	mm	
Mass of the Module	86	g	

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7.2 Mechanical Drawing of EPD Module



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8 Optical characteristics

Parameter	Conditions	Values			Units	Notes
		Min.	Typ.	Max		
White Reflectivity	White	42	45		%	-
Contrast Ratio (CR)	-	15	20		-	7-1

(T=25℃, fv = 85Hz. Measurements are made with Eye-One Pro Spectrophotometer.)

Notes:

7-1.CR = Surface Reflectance with all white pixel / Surface Reflectance with all black pixels.

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9 Handling, Safety and Environment Requirements

1. The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
2. Do not apply pressure to the EPD panel in order to prevent damaging it
3. Do not connect or disconnect the interface connector while the EPD panel is in operation
4. Do not stack the EPD panels / Modules.
5. Keep the EPD Panel / Module in the specified environment and original packing boxes when storage in order to avoid scratching and keep original performance.
6. Do not disassemble or reassemble the EPD panel
7. Use a soft dry cloth without chemicals for cleaning. Please don't press hard for cleaning because the surface of the protection sheet film is very soft and without hard coating. This behavior would make dent or scratch on protection sheet
8. Please be mindful of moisture to avoid its penetration into the EPD panel, which may cause damage during operation
9. It's low temperature operation product. Please be mindful the temperature different to make frost or dew on the surface of EPD panel. Moisture may penetrate into the EPD panel because of frost or dew on surface of EPD panel, and makes EPD panel damage.
10. High temperature, high humidity, sunlight or fluorescent light may degrade the EPD panel's performance. Please do not expose the unprotected EPD panel to high temperature, high humidity, sunlight, or fluorescent for long periods of time. Please store the EPD panel in controllable environment of warehouse and original package. Without sunlight, without condensation a temperature range of 15°C to 35°C, and humidity from 30%RH to 60%RH.
11. The EPD Panel / Module is manufactured from fragile materials such as glass and plastic, and may be broken or cracked if dropped. Please handle with care. Do not apply force such as bending or twisting to the EPD panel

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10 Reliability Test

No.	TEST	CONDITION	METHOD	REMARK
1	High-Temperature Operation	T = +50°C, RH = 30% for 240 hrs	IEC 60068-2-2Bp	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
2	Low-Temperature Operation	T = 0°C for 240 hrs	IEC 60068-2-2Ab	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
3	High-Temperature Storage	T = +70°C, RH=23% for 168hrs	IEC 60068-2-2Bp	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
4	Low-Temperature Storage	T = -25°C for 240 hrs	IEC 60068-2-1Ab	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
5	High-Temperature, High-Humidity Operation	T = +40°C, RH = 90% for 168 hrs	IEC 60068-2-3CA	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
6	High Temperature, High- Humidity Storage	T = +60°C, RH=80% for 168hrs	IEC 60068-2-3CA	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
7	Thermal Shock	1 cycle:[-25°C 30min]→[+70°C 30 min] : 50 cycles	IEC 60068-2-14	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
8	Package Vibration	1.04G, Frequency:10~500Hz Direction: X,Y,Z Duration: 1 hours in each direction	Full packed for shipment	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
9	Package Drop Impact	Drop from height of 122 cm on concrete surface. Drop sequence: 1 corner, 3edges, 6 faces One drop for each	Full packed for shipment	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
10	Electrostatic Effect (non-operating)	Machine model +/- 250V, 0Ω, 200pF	IEC 62179, IEC 62180	At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.
11	Stylus Tapping	POLYACETAL Pen: Top R0.8mm Load: 200gf;Speed:30times/min; Speed: 30times/min Total 13,500times,		At the end of the test, electrical, mechanical, and optical specifications shall be satisfied.

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11 Packing

TBD

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12 Mark and Bar Code Definition



QR code scanning content :

E133A1 Y 62 1 O4G 00001 D Z -2.10

(1) (2) (3) (4) (5) (6) (7) (8) (9)

Description:

(1) Module No. : E133A1- OPM133A1,

(2) EPL generation, Y- YZ Serial

(3) EPL batch: Printed according to the actual batch as below:

01~99	001~099	G0~G9	160~169				
A0~A9	100~109	H0~H9	170~179				
B0~B9	110~119	J0~J9	180~189				
C0~C9	120~129	K0~K9	190~199				
D0~D9	130~139	L0~L9	200~209				
E0~E9	140~149	M0~M9	210~219				
F0~F9	150~159	N0~N9	220~229				

(4) Internal code

(5) Product date year month day, printed according to the actual batch as below

M/D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Signed	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	G
Y/D	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
Signed	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W

(6) Product Serial Number

(7) Internal code

(8) Module Factory

(9) Vcom Value